Unit Cell Dispersion in CST Microwave Studio

In the tutorial, we consider numerical simulations of the dispersion diagram for a square lattice of bianisotropic dielectric resonators [1].

1 Model

The resonator consists of two coaxial cylinders with diameters $D_1 = 29.1$ mm and $D_2 = 22$ mm, and heights $h_1 = 9$ mm and $h_2 = 3$ mm, Fig. 1. The center of the resonator is located at the origin $(0, 0, 0)$. The permittivity of the resonator's material is set to 39.

For the tutorial, we use the default mesh settings, but we recommend setting the appropriate local mesh for your model (Mesh → Local properties).

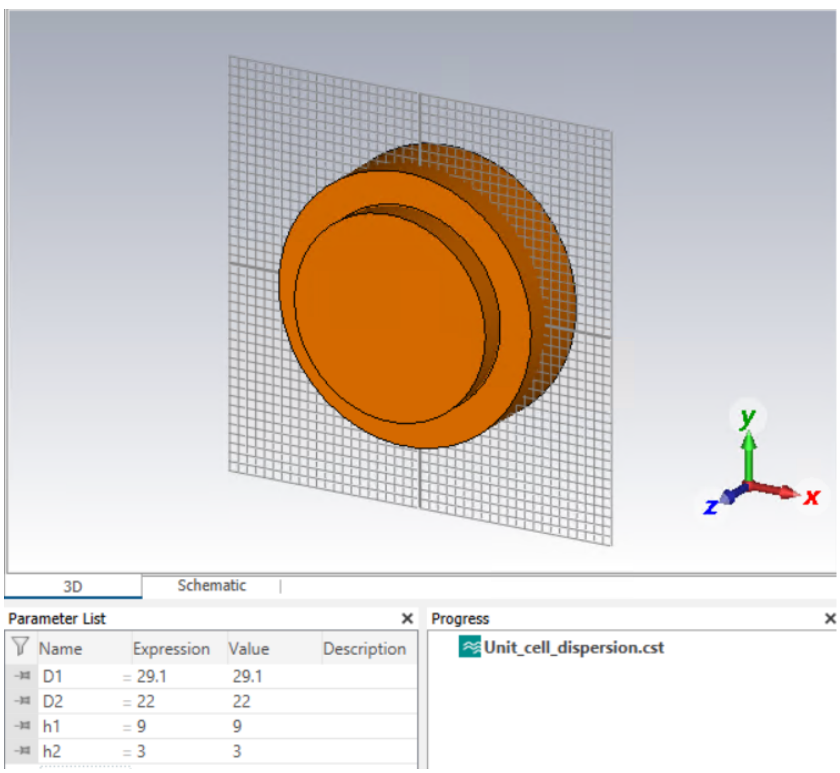

Figure 1: Numerical model of the bianisotropic dielectric resonator

2 Numerical Simulation Settings

Frequency range settings

Set the frequency range from 1 GHz to 3 GHz, Fig. 2.

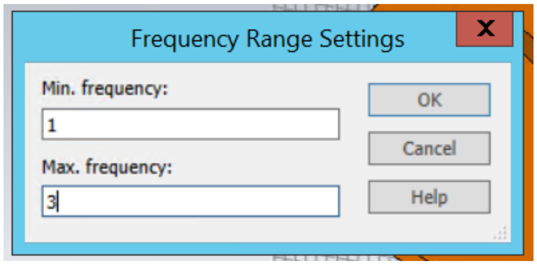

Figure 2: Frequency range settings

Background properties

Consider the square lattice of resonators in the xy-plane, with a lattice constant $a = 37.1$ mm (8 mm between the edges of the neighboring resonators). Apply a 4 mm distance between the resonator edges and the unit cell boundaries along both the x- and y-axes. The distances along the z-axis are zero, Fig. 3.

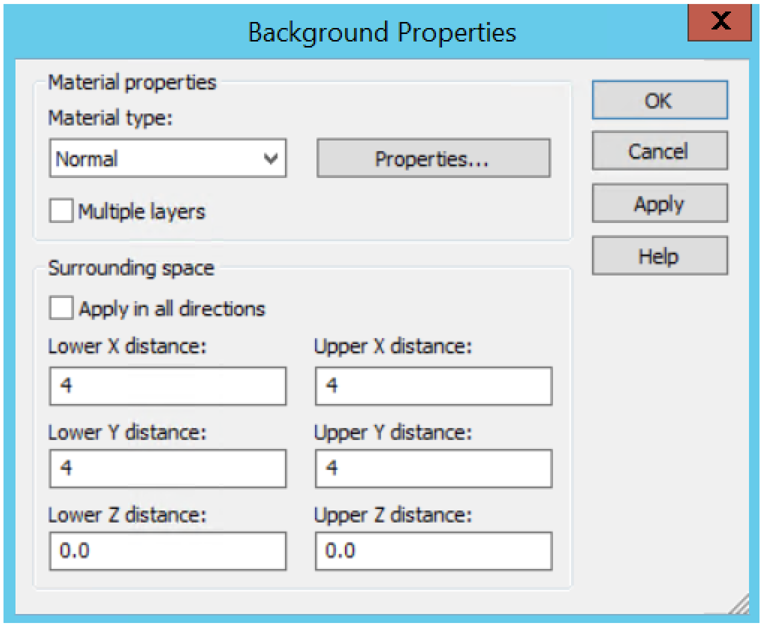

Figure 3: Background properties

Boundary Conditions

Set a periodic boundary conditions along the x- and y-axes, and an open (add space) boundary condition along the z-axis, Fig. 4.

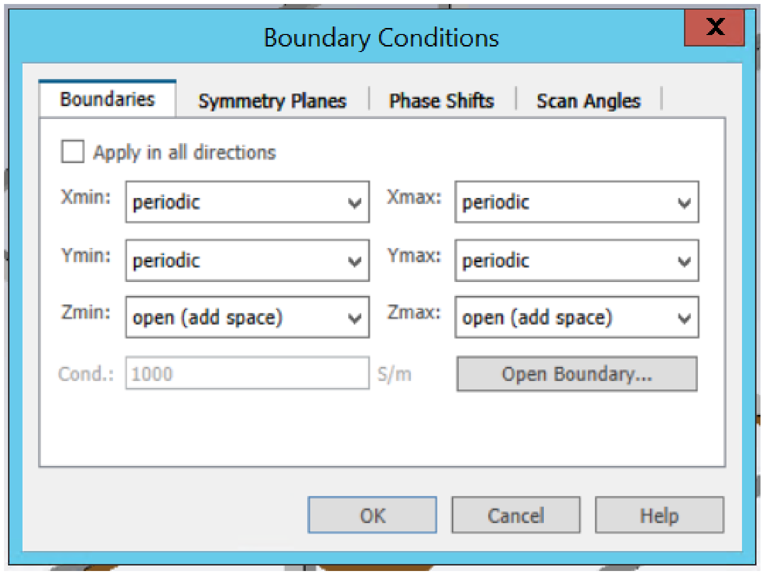

Figure 4: Boundary condition settings

Introduce new variables kx, ky, and kxy to describe phase shifts along the x- and y-axes, as shown in Fig. 5. Set the values of these variables to zero.

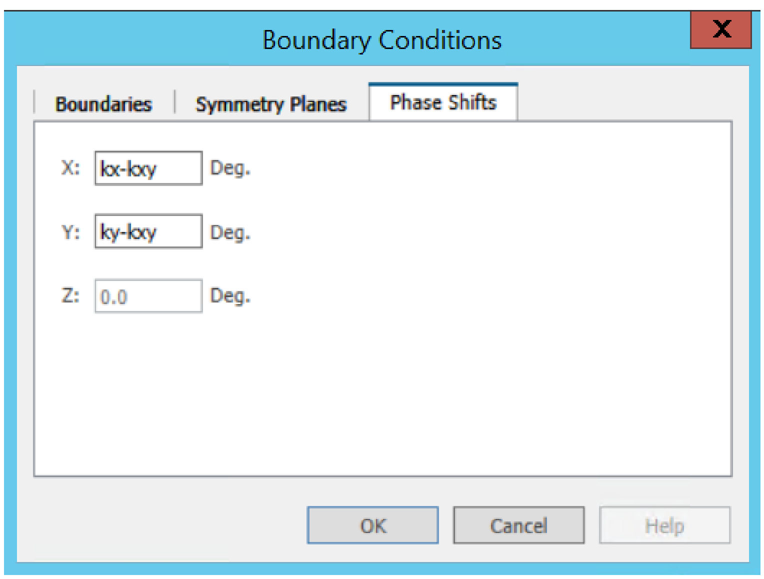

Figure 5: Phase shift settings

Solver Settings

Select the Eigenmode Solver. Specify the desired number of modes and the lowest frequency, Fig. 6. Do not use adaptive mesh refinement.

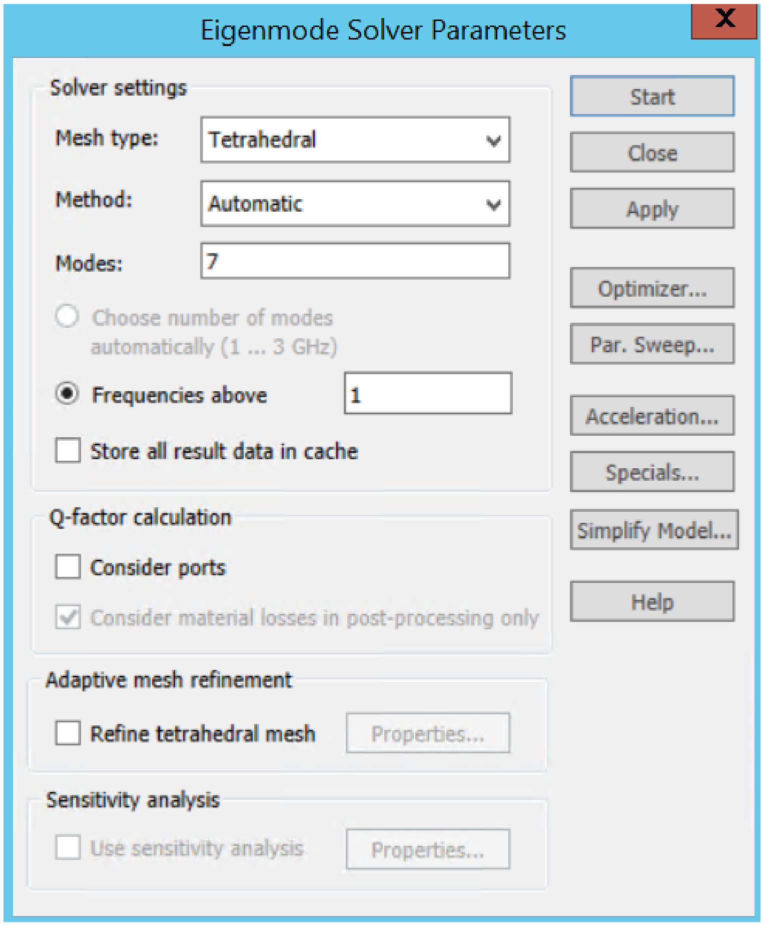

Figure 6: Eigenmode solver parameters

Parameter Sweep

To calculate the dispersion along the following trajectory of high-symmetry points with coordinates (k_x, k_y): $\Gamma(0,0) - X(\pi/a, 0) - M(\pi/a, \pi/a) - \Gamma(0,0)$, set the following sequences for the phase shifts kx, ky, and kxy:

1. Change kx from 0 to 180, while keeping $ky = 0$ and $kxy = 0$ to calculate the dispersion between Γ and X high-symmetry points.
2. Change ky from 0 to 180, while keeping $kx = 180$ and $kxy = 0$ to calculate the dispersion between X and M high-symmetry points.
3. Change kxy from 0 to 180, while keeping $kx = 180$ and $ky = 180$ to calculate the dispersion between M and Γ high-symmetry points.

In the example, the step width is set to 15, Fig. 7.

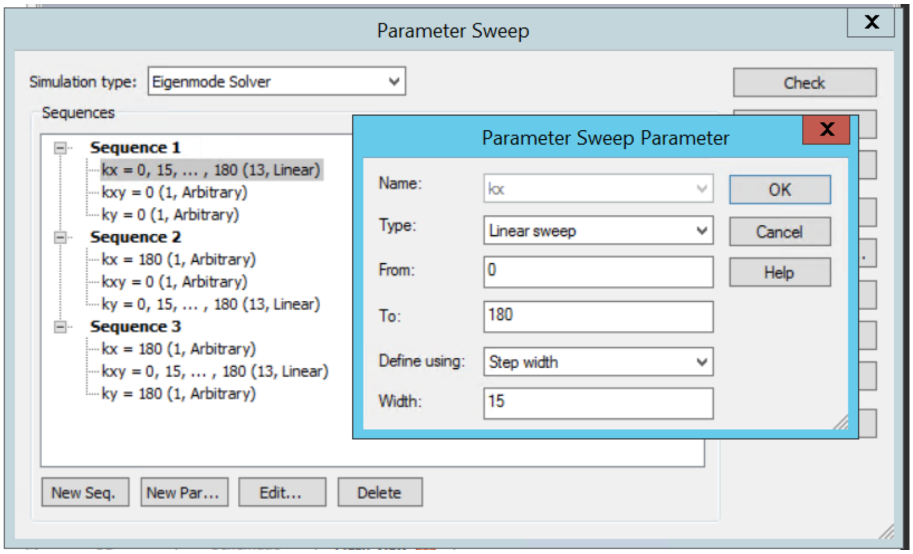

Figure 7: Parameter sweep settings

Post-Processing

The extraction of electromagnetic field profiles for each of the obtained eigenmodes is performed using VBA scripts. To add the VBA code, use Post Processing → Result Template → Misc → Run VBA. Replace the example with Hello World warning with the following code for E-field export:

```
'#Language "WWB-COM"

Option Explicit

Sub Main
    ' E-field export for modes 1..7 in xy-plane (z=7 mm)

Dim i, a, b, c, freq_str
a = kx
b = ky
c = kxy
Dim freq As Double

For i=1 To 7 STEP 1

SelectTreeItem ("2D/3D Results\Modes\Mode "&i&"\e")

freq = GetFieldFrequency()
freq_str = Replace(Format(freq, "0.000000"), ",", ".")

With ASCIIExport
    .Reset
    .FileName ("path\e_kx_"&a&"_ky_"&b&"_kxy_"&c&"_mode_"&i&"_f_"&freq_str&".txt")
    .Mode ("FixedWidth")
    .StepX (1)
    .StepY (1)
    .StepZ (1)
    .SetSubvolume(-18, 18, -18, 18, 7, 7)
    .UseSubvolume("True")
    .Execute
End With
Next
End Sub
```

or the following code for H-field:

```
'#Language "WWB-COM"

Option Explicit

Sub Main
    ' H-field export for modes 1..7 in xy-plane (z=7 mm)

Dim i, a, b, c, freq_str
a = kx
b = ky
c = kxy
Dim freq As Double

For i=1 To 7 STEP 1

SelectTreeItem ("2D/3D Results\Modes\Mode "&i&"\h")

freq = GetFieldFrequency()
freq_str = Replace(Format(freq, "0.000000"), ",", ".")

With ASCIIExport
    .Reset
    .FileName ("path\h_kx_"&a&"_ky_"&b&"_kxy_"&c&"_mode_"&i&"_f_"&freq_str&".txt")
    .Mode ("FixedWidth")
    .StepX (1)
    .StepY (1)
    .StepZ (1)
    .SetSubvolume(-18, 18, -18, 18, 7, 7)
    .UseSubvolume("True")
    .Execute
End With
Next
End Sub
```

Replace path with the required directory. To export both E- and H-field profiles for the modes, add one more Run VBA template.

Run the dispersion calculation (Parameter Sweep → Start) and wait for the result. The obtained dispersion diagram is shown in Fig. 8.

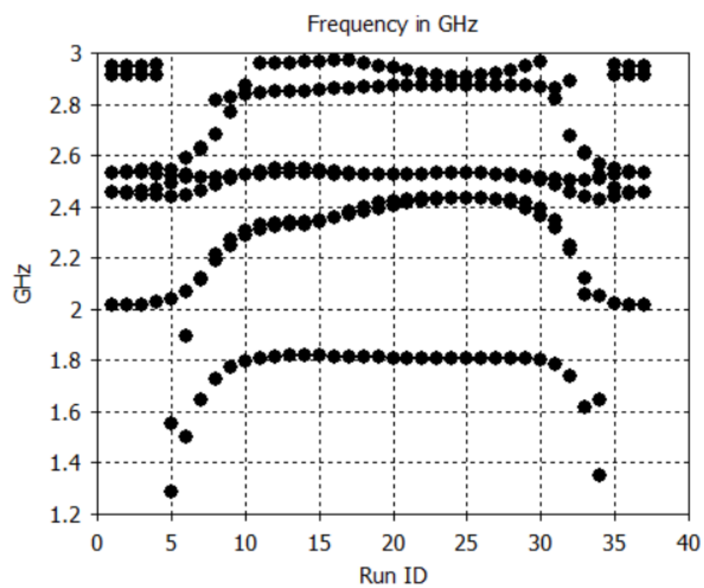

Figure 8: Calculated unit cell dispersion

References

[1] Topological edge states in all-dielectric square-lattice arrays of bianisotropic microwave resonators / Alina D Rozenblit, Georgiy D Kurganov, Dmitry V Zhirihin, Nikita A Olekhno // *Physical Review B*. — 2025. — Vol. 111, no. 8. — P. 085415. — URL: `https://link.aps.org/doi/10.1103/PhysRevB.111.085415`.